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Palaeosecular Variation of the Geomagnetic Field in the Aleutian Islands, Alaska

Douglas K. Bingham* and David B. Stone

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Sequences of late Tertiary to Quaternary lava flows at six Aleutian Island sites were sampled for palaeosecular variation (PSV) measurement using palaeomagnetic methods. Due to limitations in probable time span represented by each sequence of flows and statistical constraints, data from only two sites are considered in themselves to be reliable indicators of PSV. When all the data are combined to give greater statistical significance, an angular standard deviation of palaeofield directions of $\delta = 10{\cdot}8°$ is obtained. This is compatible with other Pacific Ocean PSV data and points to the persistence of the present secular variation low in this region at least into Quaternary time.

1. Introduction

As part of an effort to delineate the northern boundary of the postulated region of low palaeosecular variation (PSV) of the geomagnetic field centred on the Pacific Ocean (Doell & Cox 1971), sequences of Cenozoic lava flows in Alaska have been sampled and their palaeomagnetism analysed using the usual methods for PSV analysis (e.g. Cox 1969a). Preliminary results from the Aleutian Islands are presented here. Between 7 and 21 flows were sampled at six sites ranging in age from late Tertiary to Quaternary times (Fig. 1). Following Doell & Cox (1963), at least five or six 2·54-cm diameter oriented cores were taken from nearly every flow to minimize within-flow scatter; one to three 1 or 2.1 cm thick discs were cut from each core.

Several pilot samples from each site were demagnetized at successively higher 60 Hz alternating fields (AF), to determine an optimum field at which to demagnetize all the remaining samples in order to remove soft components, while at the same time retaining the stable part of the natural remanent magnetization (NRM). The field used for the overall demagnetization was chosen as that which produced the minimum scatter in the data.

Within-flow and between-flow statistics were calculated assuming a Fisher (1953) spherical normal, or Fisherian, distribution of unit vectors. The best estimate, k, of the Fisher parameter, κ, an inverse measure of dispersion, is calculated according to

$$k = \frac{N-1}{N-R} \tag{1}$$

* Now at the Institute of Earth and Planetary Physics, University of Alberta, Edmonton 7, Canada.

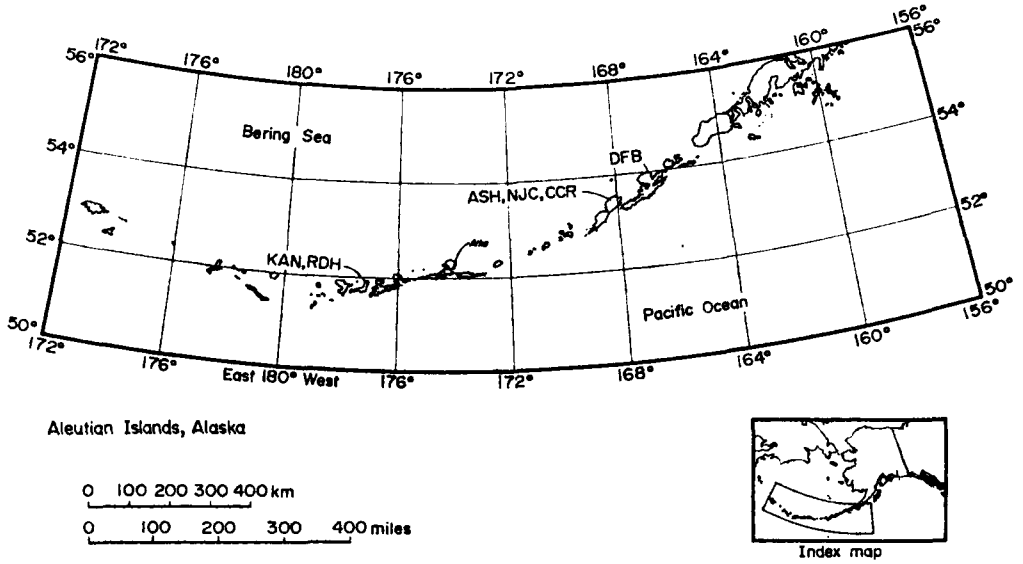


FIG. 1. Location map of palaeosecular variation measurement sites in Alaska.

where R is the length of the vector sum of N unit vectors. The angular standard deviation is given by

$$\delta = \cos^{-1}\frac{R}{N} \tag{2}$$

(Wilson 1959). The radius of the 95 per cent confidence limit of a series of vectors, α_{95}, is calculated according to Fisher (1953).

In order to determine the relative proportions of the measured between-flow dispersion ($1/k_T$) biased by within-flow dispersion ($1/k_\omega$) due to errors, and that due to true field variations of PSV ($1/k_\beta$) an analysis of variance following the method of Watson & Irving (1957), or the simpler 2-tier analysis (e.g. Creer & Sanver 1970), was run on the results from each site.

Within-flow statistical calculations were carried out on magnetic vector directions alone, while between-flow calculations (site mean data) were done on both magnetic vector directions and on virtual geomagnetic poles (VGP's) (Cox & Doell 1960). Tectonic corrections to the data were not necessary due to very gentle dips encountered.

2. Section descriptions and palaeomagnetic data

(*a*) *Kanaton Ridge, Kanaga Island (KAN).*

The geology of northern Kanaga Island in the Andreanof Island group of the Aleutian Islands is outlined by Coats (1947, 1956). Kanaton Ridge (KAN) forms the caldera rim of ancient Mount Kanaton, a broad shield volcano constructed, according to Coats, during late Tertiary and Quaternary time. However, a whole rock K–Ar date run on a sample taken a few feet above the top flow cored gave a date of only $0{\cdot}18 \pm 0{\cdot}09$ My. The section is composed of a sequence of nearly horizontal basalt and andesite flows, of which eight were sampled. The samples were taken from an eastward trending ridge just outside the ancient caldera rim, and about 4·5 km south-east of Kanaga Volcano (51° 54′ N, 182° 54′ E).

On the basis of pilot AF demagnetization tests (Fig. 2(a), and (b)), the samples were all demagnetized in peak fields of 380 Oe.

The mean directions from Kanaton are well grouped with the exception of Flow 8 (*Fig.* 2(c) and (d)). Cores showing magnetically low dips and both normal and reversed directions are present in this flow. It was decided that these cores represent two flows, the one being normal with low dip and high scatter and the other being reversed with low dip but very small scatter (Table 1). This is consistent with relative core locations in the outcrop, though no field evidence for separate flows was noted. They both have considerably lower magnetic intensity than the remaining flows. These flows have been designated 8A and 8B respectively.

Due to low magnetic dips and intensities of samples from these flows, it is presumed that they were extruded while the geomagnetic field was undergoing a reversal and therefore represent a period of unstable field behaviour (e.g. Creer & Ispir 1970); thus, they have been omitted from the statistical analysis.

The remaining seven flows have vector directions and VGP's which are closely grouped (k = 136 and 58, respectively) about the axial dipole field direction and the geographic pole, (Table 1 and Fig. 2(c) and (d)). The distribution is approximately axially symmetric. A clear serial correlation from one flow to the next is evident, indicating that perhaps these seven flows were extruded fairly rapidly and do not represent the full spectrum of secular variation, generally considered to span about 10^4 yr.

A two-tier analysis (Watson & Irving 1957) was carried out on flows 1–7. The variance ratio proved significant at the 95 per cent level so that calculated values for k_ω and k_β are given in Table 2. The number of flows is too small for a reliable measurement of PSV to be possible; nevertheless, we may take $k_\beta = 239$ as being a minimum measure (maximum k) of PSV during the time of extrusion.

Table 1

Flow mean statistical data for KAN series. Flows 8A and 8B are omitted from section mean statistics (see text)

Flow No.	N	k	δ	α_{95}	D	I	λ	ϕ	J_{nrm}
8B	5	553·6	3·1	3·3	98·2	−44·7	−25·4	283·9	855
8A	6	2·5	48·2	53·9	53·2	45·6	43·4	283·9	980
7	6	37·6	12·1	11·1	335·8	71·7	75·2	120·7	2261
6	10	41·3	12·0	7·6	345·9	65·7	80·1	74·9	1648
5	8	70·6	9·3	6·6	351·0	64·9	82·3	55·6	2828
4	7	85·7	8·1	6·6	353·9	63·5	82·1	35·6	2269
3	8	82·2	8·4	6·1	352·6	59·4	77·3	29·3	2105
2	10	51·6	10·7	6·8	11·5	61·2	77·6	319·5	1489
1	3	215·9	4·5	8·4	15·8	67·4	79·9	276·4	2337
KAN_{fld}	7	136·3	6·4	5·2	355·9	65·3	84·9	35·3	—
KAN_{vgp}	7	58·0	9·9	8·0	—	—	85·0	37·8	—

Explanation: Flows are listed in stratigraphic sequence (youngest first). Section means are listed as KAN_{fld} and KAN_{vgp}. N = number of cores per flow, or flows per site for site mean data; k, δ, α_{95} are within-flow statistical parameters calculated according to equations (1), (2) and following Fisher (1953) respectively. D and I are mean declination and inclination directions of the north seeking magnetic vector. + or − λ and ϕ are north or south geographic latitude and east longitude of corresponding VGP except for KAN_{vgp} where mean of VGP's is given. δ, α_{95}, D, I, and ϕ are all in degrees. J is the flow mean NRM value of magnetic intensity in e.m.u. $cm^{-3} \times 10^{-6}$.

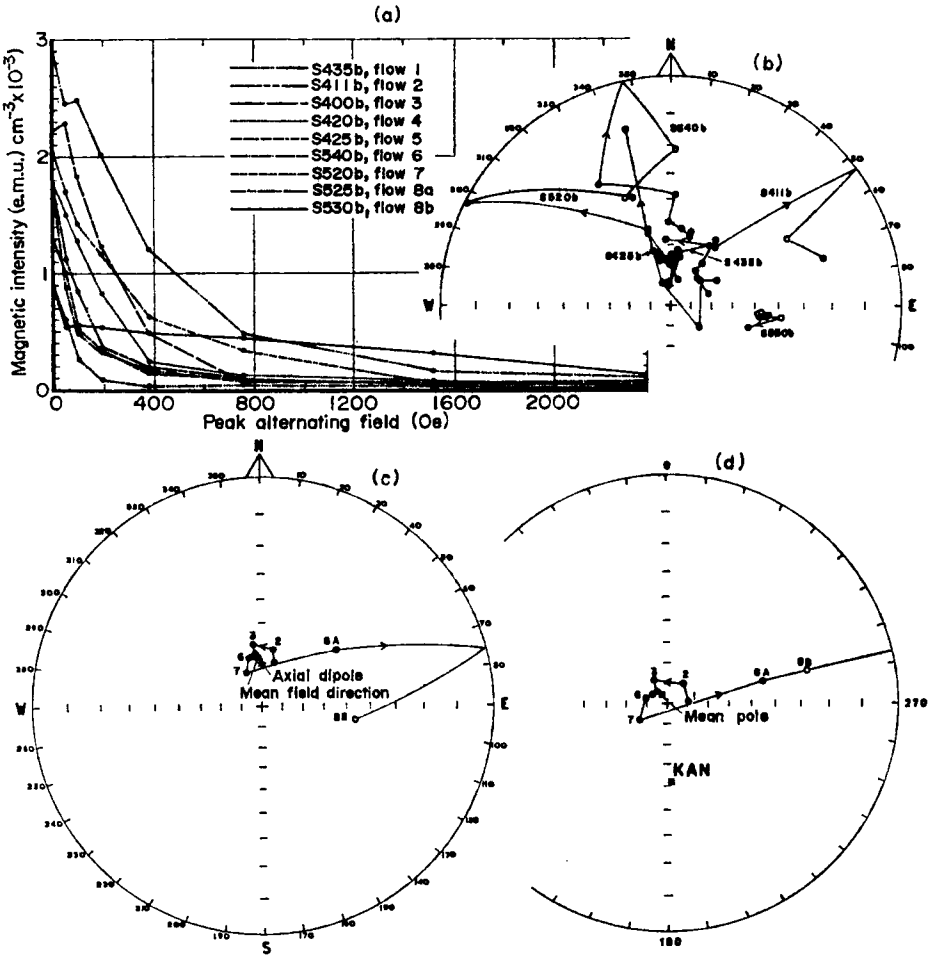


FIG. 2. Palaeomagnetic data for Kanaton Ridge, Kanaga Island (KAN). (a) Magnetic intensity and (b) vector directions of pilot samples during AF demagnetization. Arrows point toward higher fields. Sample numbers are indicated. (c) Flow mean magnetic vectors. Flow numbers are indicated; arrows point toward younger flows. (d) Corresponding flow mean VGP's; section location indicated. Mean data from Table 1. North seeking poles (north VGP's) are plotted on Wulff stereonets with solid circles indicating lower (north) hemisphere and open circles indicating upper (south) hemisphere.

(*b*) *Round Head, Kanaga Island* (*RDH*)

Round Head forms a prominent sea cliff near the easternmost extremity of Kanaga Island. It is composed of gently dipping olivine basalt flows 3–15 m thick which were extruded from a vent on the north-east side of ancient Mount Kanaton. These flows postdate the KAN sequence. The rock is fresh and typically light in colour with conspicuous large phenocrysts of dark augite (Coats 1956). A whole-rock K–Ar date on one of the lower flows yielded an age of 0·0±0·6 My and is therefore inconclusive.

Fourteen flows were sampled along the upper edge of the cliff south-east of Round Head summit at altitudes ranging from near sea level to about 230 m (near 51° 54′ N, 182°57′ E). Following AF demagnetization tests (Fig. 3(a) and (b)), each core was

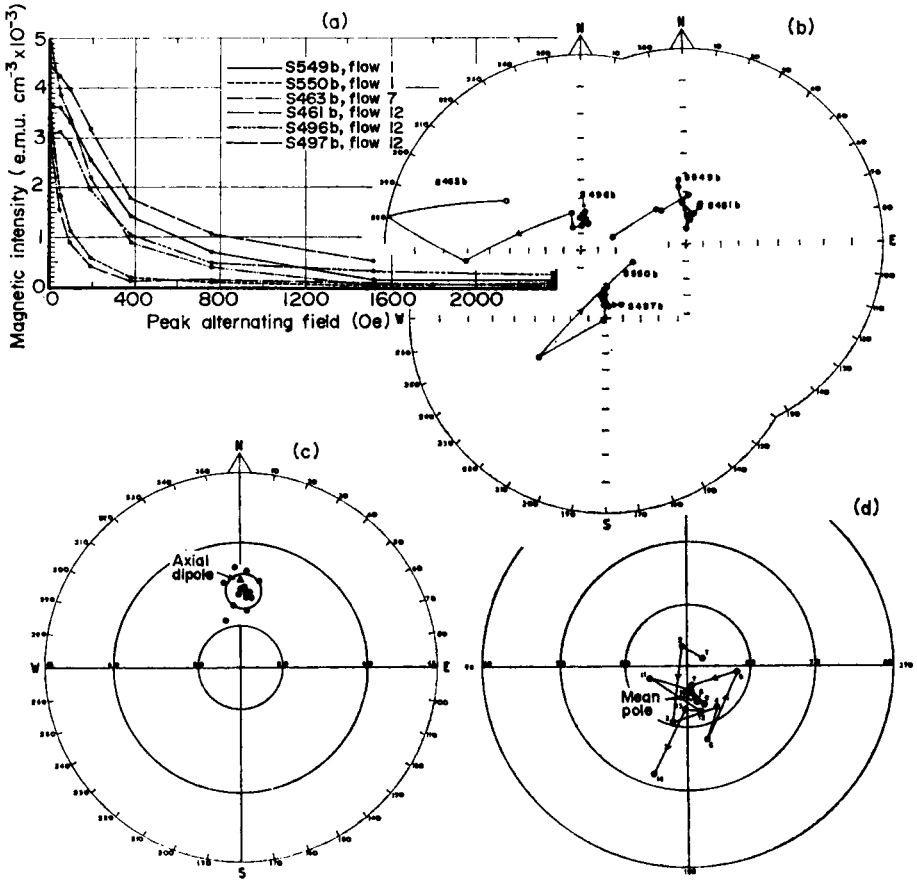


FIG. 3. Palaeomagnetic data for Round Head, Kanaga Island (RDH). (a) Magnetic intensity and (b) vector directions of pilot samples during AF demagnetization. (c) Flow mean magnetic vectors; circle of radius δ is drawn about section mean. (d) Corresponding flow mean VGP's. Note expanded stereonets. Mean data from Table 3; cf. explanation to Fig. 2 for further details.

demagnetized in 190 Oe peak field; the values thus obtained were used in the statistical analysis.

The RDH flows are all normally magnetized. All but two of the 14 flows have angular standard deviations $\delta < 10°$ and their means are tightly distributed ($\delta = 3{\cdot}9°$, $k = 401{\cdot}4$) in a roughly circular distribution about a point within 5 degrees of the axial dipole field direction (Table 3 and Fig. 3(c) and (d)). However, as the α_{95} circles of confidence for the flows have considerable overlap with each other, the significance of the separate flow mean directions as being representative of true field changes is doubtful. This is borne out by the analysis of variance (Table 2), which finds the variance ratio to be less than that which could occur randomly 5 per cent of the time. Thus it is not possible to calculate separate k_ω and k_β and all we may conclude is that $k = 401{\cdot}4$ represents the maximum dispersion (i.e. minimum k value) during the time of extrusion of the lavas.

The problem remains as to whether this value can be considered a measure of PSV or whether we must conclude that the flows were extruded too fast for PSV to be recorded. Doell (1969), arguing from the present rate of eruption and geometry

Table 2

Analysis of variance table for Aleutian Island sections, following method of Watson Irving (1957)

Section	Source	DF	SS	MS	F	$F_{5\%}$	Signif.	k_ω	k_β
KAN	between flow	12	0·2886	0·02405	2·742	1·88	yes	57·01	239·34
	within flow	90	0·7890	0·00877					
RDH	between flow	26	0·2038	0·0084	1·444	1·55	no		
	within flow	148	8·804	0·00543					
NJC	between flow	36	0·8567	0·02380	5·79	1·5	yes	121·65	142·89
	within flow	176	0·7236	0·00411					
ASH	between flow	24	2·8369	0·11832	45·16	1·63	yes	190·84	21·27
	within flow	102	0·2670	0·00262					
CCR_n	between flow	28	0·9231	0·03297	3·59	1·70	yes	54·41	153·68
	within flow	190	1·746	0·00919					
CCR_a	between flow	28	1·8387	0·06567	4·06	1·70	yes	30·92	74·53
	within flow	192	3·104	0·01617					
DFB	between flow	40	1·2732	0·03183	15·99	1·44	yes	251·26	101·33

Explanation: DF, SS and MS are degrees of freedom, sum of squares, and mean squares, respectively. F is ratio of mean squares; $F_{5\%}$ is F-ratio for 5 per cent probability level read from standard statistical tables for the listed DF's. k_ω and k_β are explained in the text. k_β is significant or not according as F is greater or less than $F_{5\%}$. For KAN series flows 8A and 8B have been omitted; likewise flow 21 in ASH series. Subscripts *n* and *a* refer to values for CCR section calculated on the bases of NRM measurements and on measurements made after 760 Oe AF demagnetization respectively.

Table 3

Flow mean statistical data for RDH series (cf. Table 1 for further explanation)

Flow No.	N	k	δ	α_{95}	$\bar{D}$	$\bar{I}$	λ	ϕ	J_{NRM}
14	5	47·9	10·5	11·2	343·0	78·4	72·2	161·7	3970·6
13	5	192·8	5·2	5·5	359·0	73·1	83·1	178·7	3292·7
12	8	541·6	3·3	2·4	3·2	73·6	82·2	195·0	3421·9
11	8	94·2	7·8	5·7	349·2	69·6	83·3	109·5	3130·1
10	8	61·9	9·7	7·1	0·8	71·3	85·9	189·0	3092·2
9	6	292·8	4·3	3·9	4·7	72·7	83·2	204·5	3096·7
8	5	86·2	7·8	8·3	2·4	72·7	83·7	194·6	2795·9
7	5	218·2	4·9	5·2	1·2	70·5	87·2	196·7	3633·7
6	9	410·7	3·8	2·5	13·5	69·0	82·0	263·3	3253·5
5	5	164·0	5·7	6·0	7·0	76·5	77·0	196·6	3129·4
4	6	27·0	14·3	31·1	9·0	73·0	81·6	216·8	2758·6
3	4	156·9	5·6	7·4	353·6	74·6	80·1	164·9	3690·6
2	7	68·0	9·1	7·4	358·7	66·1	86·5	17·6	2839·6
1	7	72·4	8·8	7·1	4·3	67·6	87·0	298·1	3742·3
RDH_{fld}	14	401·4	3·9	2·0	1·1	72·2	84·6	189·0	—
RDH_{vgp}	14	151·6	6·3	3·2	—	—	84·4	188·5	—

of Hawaiian volcanoes, deduced an accumulation rate for a series of Hawaiian lava flows. Using a similar method for the Aleutian volcanoes (Bingham 1971), one finds that the 14 RDH flows correspond to a time span of about 2300 years. The approximate nature of this calculation should be emphasized and the possibility of a some what shorter or even a much longer time span should not be excluded.

On the basis of these rather tenuous time span extrapolations, coupled with the unusually low measured scatter, we tentatively conclude that the RDH value of $k = 401{\cdot}4$ probably does not represent the average PSV very well and must be taken as an extreme lower limit (maximum k) of PSV scatter for this site.

(*c*) *New Jersey Creek, Umnak Island* (*NJC*)

The geology of Umnak Island is outlined by Chatterjee (1971) and is detailed by Byers (1959) as well as in a preliminary report by Byers *et al.* (1947). The north east end of the island is dominated by Okmok Caldera and its associated volcanics. According to Byers, Okmok Volcano was built up during late Tertiary and early

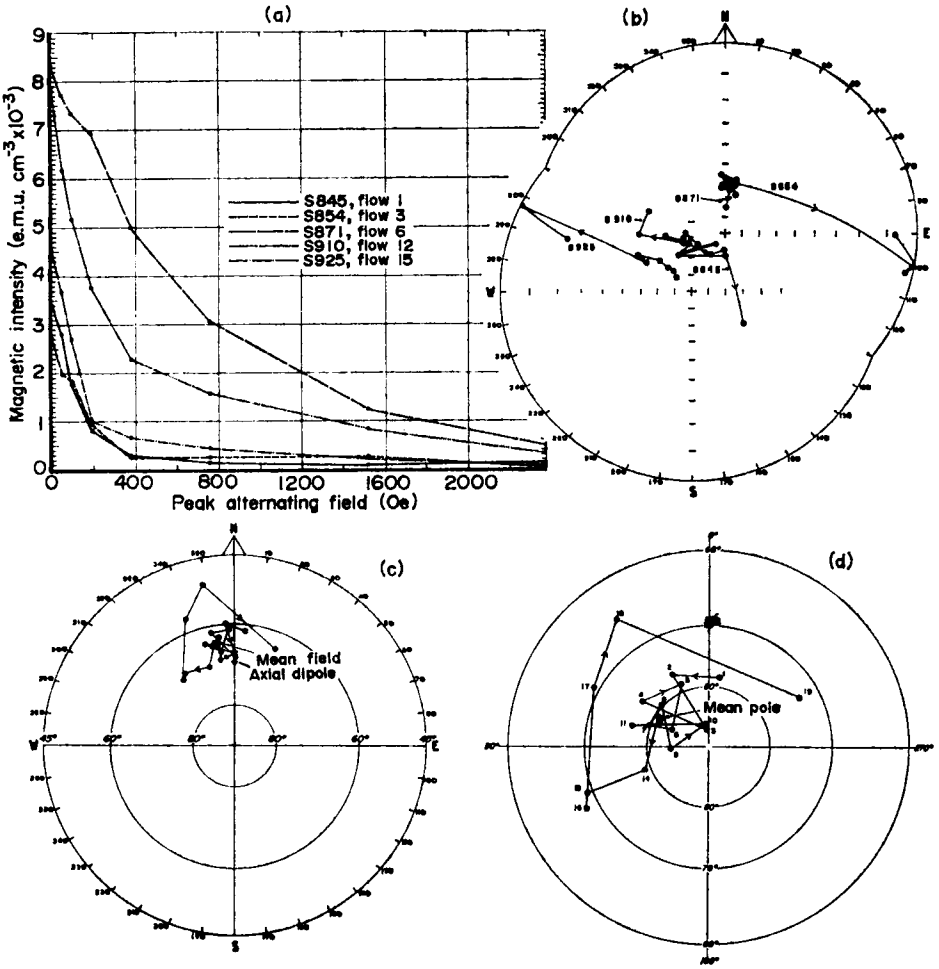


FIG. 4. Palaeomagnetic data for New Jersey Creek, Umnak Island (NJC). (a) Magnetic intensity and (b) vector directions of pilot samples during AF demagnetization. (c) Flow mean magnetic vector directions and (d) VGP's. Note expanded stereonets. Mean data from Table 4; cf. explanation to Fig. 2. for further details.

Quaternary time, and the flows sampled are among the earliest pre-caldera rocks exposed. The flows range from 6–20 m in thickness and are described as being mafic phenocryst basalt (Byers 1959). The rocks are fresh and dip very gently northwards. Their upper parts are rubbly and show evidence of considerable weathering between flow extrusions.

Nineteen flows, here called the New Jersey Creek section (NJC), were sampled from prominent exposures on a cliff about 4 km south of Ashishik Point, and 5·6 km north of the rim of Okmok Caldera (53° 32′ N, 191° 55′ E).

Each sample was demagnetized in a peak field of 475 Oe before measurement (Fig. 4(a) and (b)); the resulting values were used for the statistical analysis.

All flows were normally magnetized and had mean directions grouped in a quasi-circular distribution about a mean within 5 degrees of the axial dipole field direction (Fig. 4(c) and Table 4). The within-flow scatter is small, only two of the flows having α_{95} larger than 10° and only five more with α_{95} greater than 6°. In spite of a certain amount of overlap of α_{95} circles of confidence, the analysis of variance (Table 2) clearly indicates the calculated value of k_β to be significant. Therefore, we may take $k_\beta = 121{\cdot}7$ to represent the true dispersion of the geomagnetic field during the time the NJC flows were extruded.

Using the same extrusion rate and geometric arguments as for the RDH section, following Doell (1969), one may deduce that the 19 NJC flows probably represent a minimum time of about 15 000 years (Bingham 1971). Thus one may take $k_\beta = 121{\cdot}7$ as being a fairly reliable measure of PSV over 15000 years or so at the NJC site during later Tertiary times.

Table 4

Flow mean statistical data for NJC series. (cf. Table 1 for further explanation)

Flow No.	N	k	δ	α_{95}	D	I	λ	ϕ	J_{NRM}
19	5	586·8	3·0	3·2	23·5	64·4	73·3	298·5	5229
18	6	274·6	4·5	4·1	348·9	50·4	66·2	36·1	3006
17	6	201·0	5·2	4·7	339·2	57·5	69·0	63·2	3994
16	6	371·1	3·8	3·5	322·2	69·8	67·8	117·6	3433
15	7	134·8	6·5	5·2	325·5	68·6	69·2	111·4	4608
14	6	124·0	6·6	6·0	342·0	69·8	79·3	110·0	2807
13	6	48·6	10·6	9·7	349·0	64·8	80·6	59·8	2307
12	6	120·4	6·8	6·1	394·4	65·6	81·2	65·6	4306
11	6	251·8	4·7	4·2	343·6	64·5	77·3	74·2	8180
10	6	60·9	9·5	8·7	0·5	67·3	86·5	7·1	6314
9	6	667·6	2·9	2·6	350·2	68·8	83·9	92·5	7981
8	4	436·2	2·4	4·4	351·6	63·5	79·9	48·1	4163
7	5	174·1	5·5	5·8	347·0	65·3	79·7	70·3	6113
6	5	302·9	4·2	4·4	351·8	65·7	83·4	65·7	5405
5	5	150·8	5·9	6·3	357·0	61·4	78·8	23·4	6905
4	6	699·6	2·8	2·5	347·7	62·0	77·0	55·4	4602
3	5	52·5	10·0	10·7	0·5	68·0	87·5	4·5	2071
2	5	106·9	7·0	7·4	355·8	59·8	76·8	25·9	3293
1	6	38·1	12·0	11·0	5·5	61·5	78·5	351·1	2361
NJC_{fld}	19	122·0	7·1	3·1	350·5	64·7	80·8	57·2	
NJC_{vgp}	19	57·9	10·4	4·4	—	—	81·1	59·0	—

(d) Ashishik Basalt, Umnak Island (ASH)

Like the NJC section, the sequence of lava flows here called the Ashishik Basalt (ASH) section is part of the pre-caldera rocks of Okmok Volcano. Both sections are part of what Byers (1959) calls Ashishik Basalt. The ASH section is made up of aphyric and feldspathic basalt flows 3–25 m thick (Byers 1959). Well weathered layers between flows made differentiating one from another in the field relatively easy.

Two K–Ar whole rock age determinations of a rock from the topmost flow sampled yielded dates of $1{\cdot}7 \pm 0{\cdot}2$ and $2{\cdot}1 \pm 0{\cdot}2$ My. The ASH flows probably post-date the NJC flows although some interbedding does occur elsewhere in the section (Byers 1959).

Fourteen flows were samples from a north-east facing exposure 2·4 km south-west of Ashishik Point and about 3·2 km north-north-west of the NJC section (53° 32′ N, 191° 54′ E). The flows are numbered 8 to 21 with 21 being the top flow. Each core was AF demagnetized at 380 Oe before measurement (Fig. 5(a) and (b)).

The bottom 13 of the 14 ASH flows were found to be reversely magnetized with small within-flow scatter (all but one $\alpha_{95} < 10°$), while the topmost flow (flow 21) is weakly normally magnetized and displays very large scatter ($\alpha_{95} = 84{\cdot}5°$); this flow is ignored in the site mean statistical analysis. Complete data are given in Table 5 and Fig. 5(c) and (d). The two-tier analysis (Table 2) indicates that the between-flow scatter has considerable relevance and is not caused by within-flow scatter. The flow mean field directions (Fig. 5(c)) are fairly widely scattered ($k = 20{\cdot}8$) in a quasi-oval distribution. The site mean lies about 30° away from the axial dipole field direction.

As done previously, we may extrapolate from Doell (1969) to find that the ASH sequence represents about 10 000 years, thus spanning the full PSV spectrum. However, due to small N, the ovate distribution, and the mean pole position far removed from the axial pole, the value of $k = 20{\cdot}8$ cannot be considered a good representation of PSV during late Tertiary times.

Table 5

Flow mean statistical data for ASH series. Flow 21 *is not included in site mean data (cf. Table* 1 *for further explanation)*

Flow No.	N	k	δ	α_{95}	$\bar{D}$	$\bar{I}$	λ	ϕ	J_{NRM}
21	5	1·8	56·5	84·5	84·5247·0	83·8	47·5	175·2	2341
20	5	925·4	2·4	2·5	207·0	−66·5	−72·6	106·1	8477
19	5	465·5	3·4	3·6	199·6	−69·8	−78·4	93·2	7176
18	5	251·1	4·6	4·8	156·8	−54·4	−65·2	242·4	2964
17	5	177·6	5·4	5·8	191·5	−54·4	−69·7	163·7	5898
16	5	323·6	4·0	4·3	121·4	−72·3	−57·6	312·9	2452
15	5	103·8	7·1	7·5	185·3	−79·8	−73·1	18·1	4238
14	5	456·2	3·4	3·6	255·9	−82·2	−54·5	38·1	6584
13	4	76·0	8·1	10·6	5·6	−83·8	−41·3	10·3	7246
12	5	408·8	3·6	3·8	330·7	−80·1	−36·1	23·4	3761
11	5	192·1	5·2	5·5	4·7	−78·3	−31·1	9·8	9378
10	5	263·2	4·5	4·7	351·6	−78·7	−31·9	15·6	14089
9	5	531·4	3·1	3·3	245·6	−78·0	−56·7	52·5	8286
8	5	64·7	9·0	9·6	233·3	−73·6	−60·8	68·6	3071
ASH_{fld}	13	20·8	17·1	9·3	198·1	−81·5	−68·9	26·2	—
ASH_{vgp}	13	7·3	29·1	16·4	—	—	−66·2	25·7	—

(e) Crater Creek, Umnak Island (CCR)

A sequence of 15 lava flows were samples on the north-west side of Crater Creek gorge where it cuts through the wall of Okmok Caldera. The sampling site is about 0·8 km outside the line of the Caldera rim (53° 28′ N, 191° 55′ E). The Crater Creek basalt, defined by Byers *et al.* (1957), is the youngest pre-caldera unit of Okmok Volcano, and Byers (1959) lists these basalts as being Pleistocene to Recent in age. Whole rock K–Ar age determinations on rocks from one of the upper flows (flow No. 10) yielded an age of 0·0±0·15 My so Byer's estimate cannot be improved upon.

Chatterjee (1971) conducted mineralogical and magnetic studies on the same samples as used here in an attempt to understand the rather unusual palaeomagnetic results. The flows range from tholeiites to olivine tholeiites, and the opaques responsible for the NRM, are fine grained and restricted to the groundmass.

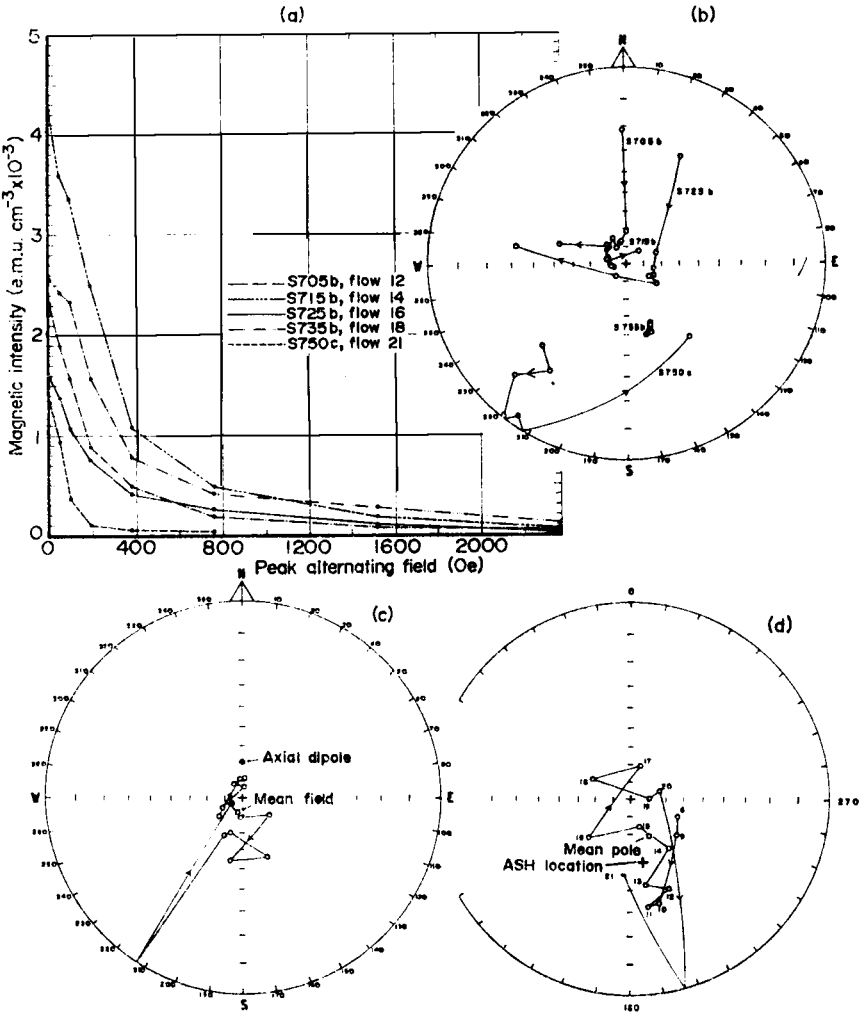


FIG. 5. Palaeomagnetic data for Ashishik Basalt, Umnak Island (ASH). (a) Magnetic intensity and (b) vector directions of pilot samples during AF demagnetization. (c) Flow mean magnetic vector directions and (d) VGP's. Mean data from Table 5; cf. explanation to Fig. 2 for further details.

Chatterjee concluded on the basis of his mineralogical studies, thermal demagnetization results and X-ray diffraction analysis, that most of the flows had a relatively stable but more or less random chemical remanent magnetization (CRM) component residing in very fine grains of haematite and/or maghemite in the ground mass which was not easily removed by standard AF demagnetization techniques.

A 760 Oe AF demagnetization was chosen as the optimum field (Fig. 6(a) and (b)) and each core was demagnetized at this field before measurement. On comparison of the demagnetized results with the NRM results it was found that in a number of flows the demagnetization increased the within-flow scatter of field vectors. For this

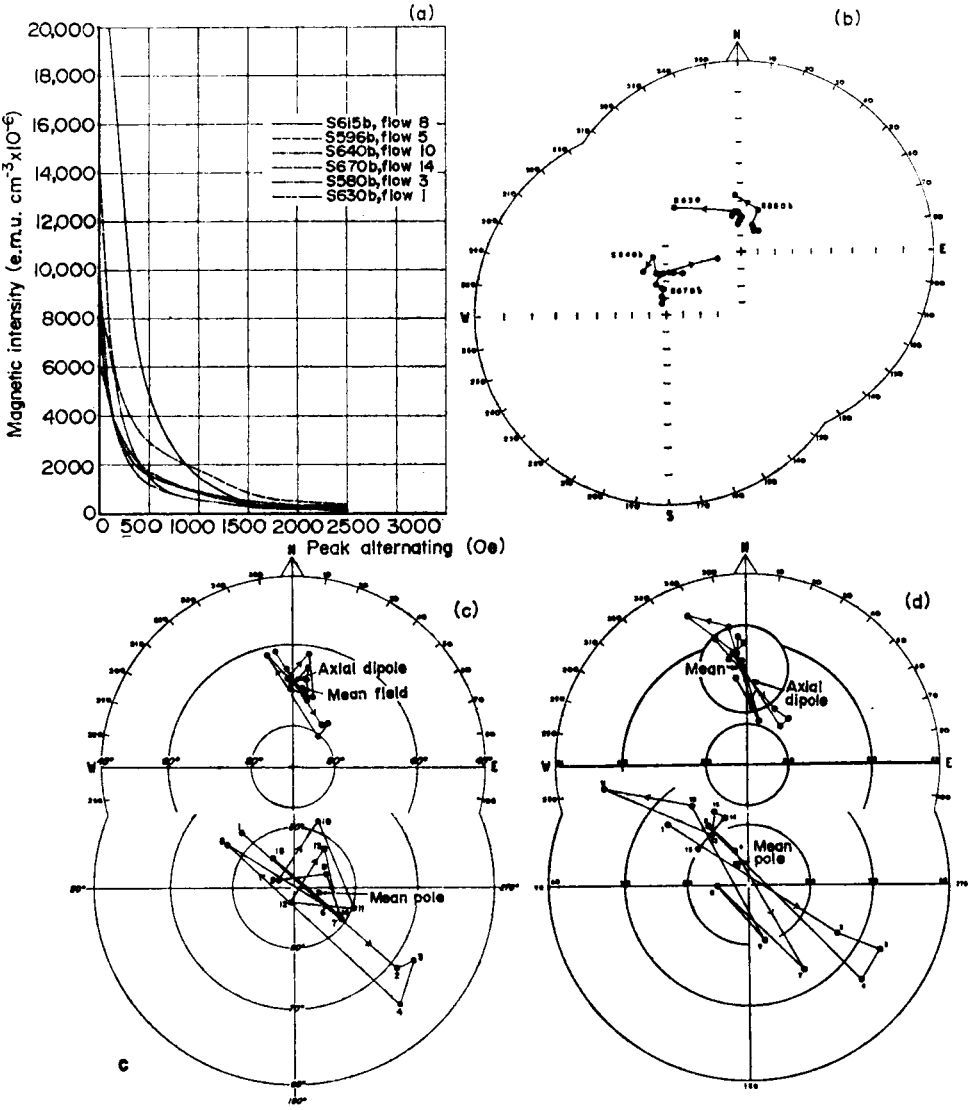


FIG. 6. Palaeomagnetic data for Crater Creek, Umnak Island (CCR). (a) Magnetic intensity (Chatterjee 1971) and (b) vector directions of pilot samples during AF demagnetization. (c) Flow mean magnetic vector directions (upper) and VGP's (lower) using NRM data. (d) Flow mean magnetic vector directions (upper and VGP's (lower) for 760 Oe. AF demagnetization. Circle of radius δ drawn about section mean vector direction. Note expanded stereonets. Mean data from Table 6; cf. explanation to Fig. 2 for further details.

Table 6

Flow mean statistical data for CCR series. Subscripts n and a indicate values calculated from NRM and 760 Oe AF demagnetized data respectively (cf. Table 1 for further explanation)

Flow No.	N_n	N_a	k_n	k_a	δ_n	δ_a	α_{95n}	α_{95a}	$\bar{D}_n$	I_n	$\bar{D}_a$	I_a	λ_a	ϕ_n	λ_a	ϕ_a
15	7	7	186·6	452·2	5·5	3·5	4·4	2·8	356·8	65·9	350·4	64·0	84·3	34·1	80·1	54·3
14	6	6	44·5	13·0	11·1	20·6	10·2	19·3	11·6	73·0	358·1	60·7	81·8	238·8	78·1	18·7
13	8	8	52·3	18·9	10·5	17·5	7·7	13·1	9·2	65·6	356·2	59·5	81·9	322·3	76·6	24·6
12	5	6	22·6	7·8	15·3	26·7	16·5	25·6	358·4	71·0	354·6	63·4	87·8	167·8	80·8	36·7
11	8	8	24·3	10·8	15·4	23·2	11·5	17·6	16·4	72·5	339·4	51·9	79·8	250·1	64·4	55·3
10	11	11	53·5	29·3	10·6	14·3	6·3	8·6	8·4	62·2	352·1	57·5	78·6	339·8	73·7	34·6
9	7	7	12·9	106·1	21·0	7·3	17·5	5·9	356·8	68·6	1·6	75·8	87·5	64·1	80·3	196·1
8	6	6	182·8	170·0	5·5	5·7	5·0	5·2	8·7	68·7	353·4	69·0	84·6	293·3	85·9	90·9
7	6	6	223·6	31·8	5·0	13·2	4·5	12·1	12·3	73·5	16·4	79·0	81·0	236·1	73·0	212·4
6	8	8	85·5	48·8	8·2	10·9	6·0	8·0	7·9	72·1	358·2	65·4	84·2	239·7	84·0	23·3
5	5	5	145·8	42·1	6·0	11·2	6·4	11·9	347·6	62·3	354·4	61·5	77·2	56·7	78·6	33·1
4	7	7	120·4	200·5	6·8	5·3	5·5	4·3	38·6	80·2	40·2	78·5	65·7	221·7	66·3	229·2
3	9	9	270·7	256·4	4·6	4·8	3·1	3·2	38·6	76·3	42·2	75·1	67·9	238·4	66·2	243·9
2	9	9	287·0	61·6	4·5	9·7	3·0	6·6	34·0	77·5	26·5	75·0	69·2	231·7	73·7	240·5
1	8	8	84·4	186·8	8·3	5·6	6·1	4·1	351·6	61·6	346·2	59·2	77·9	42·8	73·6	52·2
CCR_{fld}	15	15	115·7	58·2	7·3	10·3	3·6	5·1	6·9	70·6	358·0	67·1	85·8	259·9	86·2	31·7
CCR_{vgp}	15	15	45·6	26·0	11·6	15·4	5·7	7·6	—	—	—	—	85·2	254·8	87·6	33·4

reason, full statistical calculations on both the NRM and AF demagnetized results have been carried out (Tables 2 and 6).

The distribution of the flow mean directions and VGP's of the CCR section is unusual. All flows are normally magnetized and their VGP's display a clearly elongate, almost linear, distribution centred within 3° of the geographic pole and oriented roughly along the 45° and 225° E longitude meridians (Fig. 6(c) and (d)). The elongate quality of the distribution is somewhat less pronounced for the NRM than it is for the AF demagnetized data. This distribution is difficult to explain. On the basis of susceptibility anistropy measurements, Chatterjee (1971) concludes that susceptibility anisotropy in a preferred direction is not a possible cause for the distribution. Tectonic movements also seem unlikely because of their necessary magnitude and lack of evidence for such movements. The α_{95} circles of confidence are admittedly rather large, and some of the elongate distribution may be due to chance, but it seems that a large part of it must be due to field changes during the time of extrusion of the lava flows. This conclusion is supported by the analysis of variance of both NRM and treated data (Table 2). Nur & Helsley (1971) report a somewhat similar distribution of VGP's from Tertiary lava flows in Israel; in their case, reversed and normal VGP's are included and they line up roughly along the 135° and 315° meridians. Hope (1957) suggests that during very recent times (the last several thousand years) PSV may actually be in large part due to just such an oscillation of the north geomagnetic pole along a roughly 90–270° E axis centred close to the geographic pole. Such a pattern may have occasionally prevailed along different meridians at earlier times.

With such an elongate distribution the usual Fisher (1953) statistics must be viewed with caution. For comparison's sake, all the statistical parameters for both NRM and AF demagnetized data have been calculated. Because of their lower within-flow scatter, the NRM values seem to hold the greatest promise of being used in analyses hereafter. In fact the difference between NRM and AF mean data is not great.

Using the same method of calculating the time span as used previously, we find that the CCR section might represent about 4000 years as a minimum value. This hypothesis is borne out by weathering between flows.

Because of the short time interval represented and the non-Fisherian distribution it must be concluded that the NRM value of $k_\beta = 153{\cdot}7$ is a relatively poor value for PSV for this site.

(*f*) *Driftwood Bay, Unalaska Island (DFB)*

The geology of Unalaska Island is outlined by Drewes *et al.* (1961). They delineate the Makushin Volcanics, in which the section sampled lies, as being a thick pile of little deformed volcanic rocks associated with present day Makushin Volcano on the north-west part of the island. The formation consists primarily of basalt and andesite flows 2–15 m thick and dates from middle to late Pleistocene (Drewes *et al.* 1961).

The sampled section (DFB) is a series of 21 lava flows in a 300 m high cliff 0·2 km west of and below the Driftwood Bay White Alice and DEW Line station on the north-west shore of Unalaska Island and 4 km west of Driftwood Bay itself (53° 58′ N, 193° 16′ E). The flows appear to be quite recent and the rock fresh and unweathered, and probably date from late Pleistocene or early Holocene times. There is considerable thickness of grass-covered rubble between most of them.

Because of the very stable directions that prevailed during step-wise demagnetization (Fig. 7(a) and (b)) and the tight within-flow grouping of the NRM values, it was decided to dispense with any 'blanket' AF demagnetizing treatment and use the NRM data for analysis. The additional scatter generated by soft components not removed is thought to be negligible.

The DFB flows are all normally magnetized. The scatter within flows is small; only three flows have δ larger than 5° (Table 7). The two-tier analysis (Table 2) shows that the between-flow scatter is quite significant relative to the within-flow scatter, thus giving the calculated values of k_ω and k_β good reliability. The flow mean field directions have a small slightly elongate distribution ($k = 94{\cdot}8$), centred within a degree of the axial dipole field direction (Fig. 7(c)).

One can extrapolate, as previously, from Doell's (1969) time span calculations and deduce that the DFB section may represent about 16 000 years. Recognizing the inherent dangers of such an extrapolation, it nonetheless is reasonable to assume that the DFB section represents at least 10 000 years and take $k_\beta = 101{\cdot}33$ as being a reliable indicator of PSV at this location during the late Pleistocene.

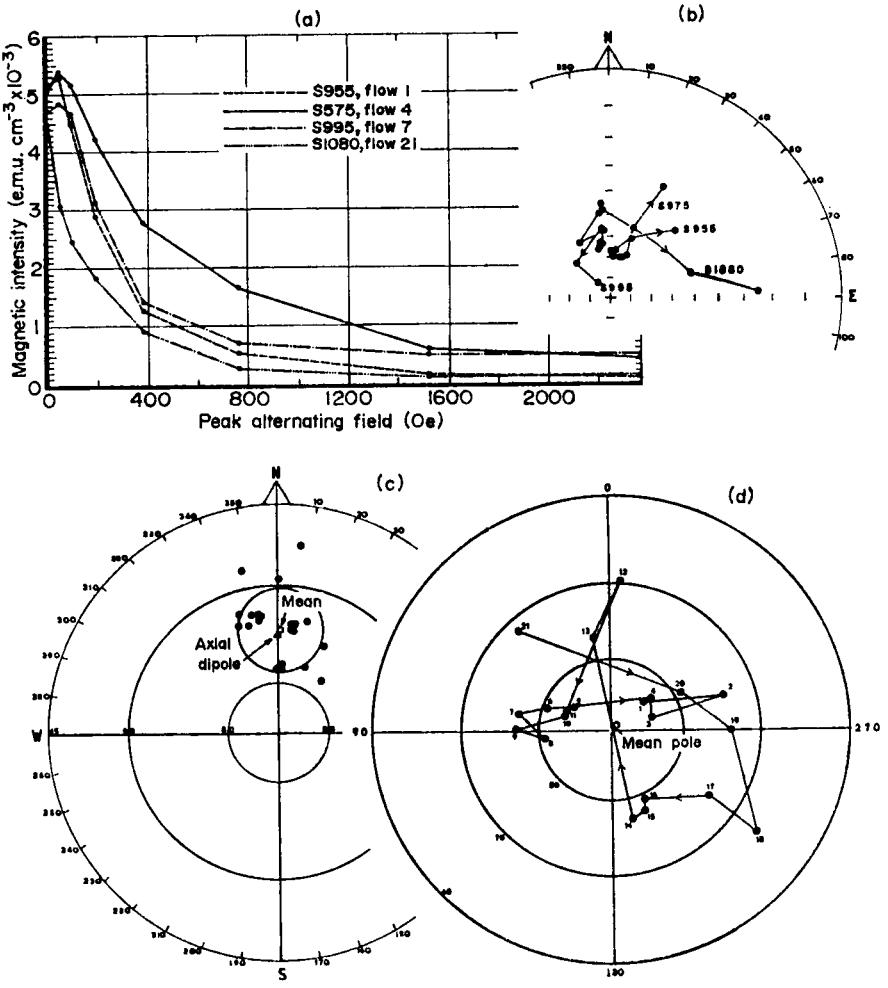


FIG. 7. Palaeomagnetic data for Driftwood Bay, Unalaska Island (DFB). (a) Magnetic intensity and (b) vector directions of pilot samples during AF demagnetization. (c) Flow mean magnetic vectors. Circle of radius δ is drawn about section mean. (d) Corresponding flow mean VGP's. Note expanded stereonets. Mean data from Table 7; cf. explanation to Fig. 2 for further details.

3. Aleutian Island mean secular variation

Of the six Aleutian Island sites discussed above, only two sites appear to be such as to give reliable readings for secular variation on their own; these are the NJC and DFB sites. The other PSV determinations are of dubious value because of either grossly non-Fisherian distribution or short time spans being represented. In order to obtain a more representative indication of late Tertiary–Quaternary PSV the data from all the flows at the six Aleutian Island sites have been combined.

If field direction statistics are to be used for the combined data as previously, rather than VGP statistics, it becomes necessary to make the assumption that the differences in field directions at different sites, corresponding to the same VGP's, are small compared with PSV effects on field directions. This is a reasonable assumption if the sites are situated at approximately the same (intermediate) palaeolatitude. The KAN, RDH, NJC, ASH, CCR, and DFB sites have a spread in latitude of only 2·1° and the longitudinal spread is 10·2°. All the flows used, except for those at the ASH site, have VGP's near enough to the present geographic pole so that their corresponding palaeolatitudes differ by approximately the same amounts as their present latitudes. The ASH flows having low latitude VGP's are fully compatible with the above assumption since this happens at only *one* site; one can thus consider all the other data 'transferred' to the ASH site.

An additional site has been included in the Aleutian Island statistics. Twelve cores were taken from one lava flow on the north-east side of Atka Harbor on Atka Island (52° 14′ N, 185° 50′ E). The unit sampled lies on the lower slopes of Mount

Table 7

Flow mean statistical data for DFB series based on NRM values (cf. Table 1 for further explanation)

Flow No.		N	k	δ	α_{95}	D	I	ϕ	J_{NRM}
1	6	176·3	5·6	5·1	7·5	67·9	84·5	314·4	5447
2	6	89·1	7·8	7·1	8·4	69·6	85·0	286·8	3705
3	6	223·3	5·0	4·5	9·6	69·3	84·2	290·2	4245
4	6	1192·6	2·1	1·9	9·0	68·0	83·8	307·0	2741
5	6	417·7	3·6	3·3	353·3	66·8	83·9	58·3	3959
6	6	235·9	4·8	4·4	348·5	66·1	81·0	70·6	4740
7	6	322·8	4·1	3·7	342·9	65·4	77·4	78·9	10182
8	6	617·0	3·0	2·7	345·0	68·6	80·8	96·0	7601
9	6	432·9	3·6	3·2	340·9	67·1	77·5	91·2	4430
10	6	284·2	4·4	4·0	351·3	67·4	83·5	71·7	2427
11	6	845·5	2·5	2·3	351·0	67·0	82·9	68·6	4423
12	6	224·9	4·9	4·5	6·9	53·9	69·8	356·3	4520
13	6	895·9	2·5	2·2	0·9	60·0	76·9	10·2	5969
14	6	290·3	4·3	3·9	359·6	77·8	77·4	192·3	11060
15	6	863·9	2·5	2·3	4·8	77·4	77·8	202·4	6312
16	7	286·6	4·4	3·6	4·9	76·8	78·8	203·9	3345
17	6	234·7	4·8	4·4	23·8	76·5	74·1	232·7	3424
18	6	55·1	10·0	9·1	41·3	77·9	66·5	233·9	15054
19	6	441·1	3·5	3·2	28·1	70·4	73·7	269·8	11492
20	6	462·3	3·4	3·1	15·8	67·1	79·5	299·6	5673
21	6	373·6	3·8	3·5	347·2	57·2	71·6	46·9	2213
DFB_{Tid}	21	94·8	8·1	3·3	0·9	69·1	88·5	351·1	—
DFB_{VGP}	21	40·7	12·4	5·0	—	—	89·0	322·5	—

Table 8

Site mean statistical data for Atka (one flow only). NRM values used (cf. Table 1 for further explanation)

	N	k	λ	α_{95}	D	I	λ	ϕ
$Atka_{fld}$	12	17·1	18·4	10·8	342·8	64·5	77·4	75·1
$Atka_{vgp}$	12	22·5	16·4	9·4	—	—	79·4	84·1

Korovin, a currently active (Coats 1950) Quaternary volcano. The basaltic flow appeared to be very fresh but not historic. Two K–Ar whole rock dates run on a hand sample from the flow yielded inconclusive dates of $0{\cdot}19 \pm 0{\cdot}38$ and $0{\cdot}24 \pm 0{\cdot}31$ My. As the NRM results appeared stable the samples were not demagnetized. The statistical results for Atka are given in Table 8.

The results of treating the six Aleutian PSV sites together with Atka are shown in Table 9 along with a summary of the Aleutian Island data discussed above. Both palaeofield direction and VGP data are given. A composite plot of all the palaeofield data is shown in Figure 8. With 90 flows, and certainly more than 10^4 years represented, we may take these values ($k = 56$, $\delta = 10{\cdot}8°$) as good estimates of PSV during the late Tertiary and Quaternary period in the Aleutian Islands region.

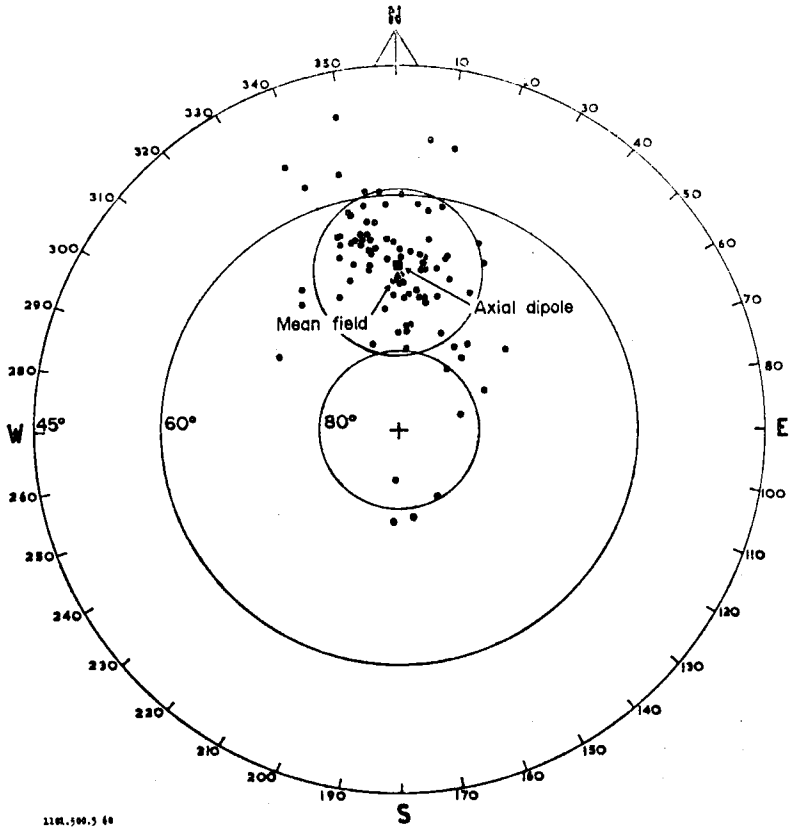


FIG. 8. Flow mean palaeofield directions for all Aleutian Island PSV sections. A circle of radius δ is drawn about the mean field direction. Note expanded stereonet.

Table 9

Summary of Aleutian Island statistical data. Lat. and Long. are latitude and longitude of flow sequence.

Age is given in million years; T-Q implies Tertiary to Quaternary age.

Statistical quantities as defined in Tables 1 and 2 are calculated for both palaeofield directions and VGP's.

					Palaeofield directions							VGP's				
	Lat.	Long.	Age	N	k_T	k_ω	k_β	δ	α_{95}	$\bar{D}$	$\bar{I}$	k_T	δ	α_{95}	λ	ϕ
Driftwood Bay (DFB)	54·0	193·1	<1	21	94·8	251·8	101·3	8·1	3·3	0·9	69·1	40·7	12·4	5·0	89·0	322·5
Crater Creek (CCR)	53·5	191·9	<·16	15	115·7	54·5	153·7	7·3	3·6	6·9	70·6	45·6	11·6	5·7	85·2	254·8
New Jersey Creek (NJC)	53·5	191·9	T−Q	19	122·0	121·7	142·9	7·1	3·1	350·5	64·7	57·9	10·4	4·4	81·1	59·0
Ashishik (ASH)	53·5	191·9	<1·9±·2	13	20·8	190·8	21·3	17·1	9·3	198·1	−81·5	7·3	29·1	16·4	−66·2	25·7
Kanaton Ridge (KAN)	51·9	182·9	<·18±·09	7	136·3	57·0	239·3	6·4	5·2	355·9	65·3	58·0	9·9	8·0	85·0	37·8
Round Head (RDH)	51·9	182·9	<0·6	14	401·4	—	—	3·9	2·0	1·1	72·2	151·6	6·3	3·2	84·4	188·5
Aleutian mean[1]	53	188	T−Q	90	56·0	—	56[2]	10·8	2·0	359·4	70·4	19·5	18·3	3·5	88·0	197·8

[1]Includes six sequences above in addition to Atka from Table 8.

[2]Calculated according to simplified 2-tier analysis $\delta_\beta^2 = \delta^2 - (\delta_\omega^2/N)$, where δ_ω is the RMS within-flow dispersion and using $\delta = (81/\sqrt{k})$ (Runcorn 1957).

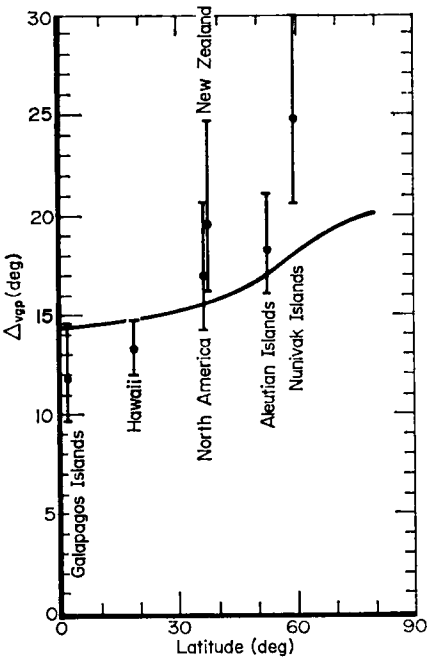


FIG. 9. Angular standard deviation of VGP's from the Pacific Ocean region. Solid curve is Cox's (1962) secular variation model (redrawn from Cox 1969b).

4. Conclusions

In Fig. 9, from Cox (1969a), the Aleutian mean value of $\delta_{VGP} = 10{\cdot}3°$ is plotted against geographic latitude along with Cox's PSV data from the Pacific region. Also included in the diagram is a curve based on Cox's (1962) secular variation model which includes both dipole wobble and non-dipole field components. Within the limits of the error bars (Cox 1969b), the data are consistent with this model except for Nunivak and New Zealand. However, a stronger latitude dependence of PSV than predicted by the model is indicated; specifically, low latitude values fall below the curve while high latitude ones fall above. It is perhaps noteworthy that the Aleutian value falls below the curve defined by the other points.

Thus PSV in the Aleutian Islands is consistent with other data from the Pacific area of low PSV. Therefore, it may be tentatively concluded that this region of low PSV has persisted since early Quaternary times, and perhaps since late Pliocene, and included the Aleutian Islands within its boundaries.

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Geophysical Institute
University of Alaska
College
Alaska 99701

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